A Secure Blockchain-Based Voting System Using Face Recognition

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Introduction

Voting is a fundamental process in democratic systems, including academic institutions and organizations, where fairness, transparency, and security are essential. However, traditional voting methods, whether paper-based or centralized electronic systems, often face challenges such as identity fraud, vote tampering, and lack of transparency. These issues can reduce trust in the voting process and create opportunities for manipulation. With the rapid advancement of digital technologies, there is a growing need for a secure and reliable electronic voting system that ensures proper voter authentication and protects vote integrity. Blockchain technology offers a decentralized and tamper-proof platform for storing voting records, while biometric authentication methods like face recognition, along with OTP verification, enhance voter identity validation. This project focuses on developing a blockchain-based voting system that combines secure authentication mechanisms with immutable vote storage to create a transparent and trustworthy voting environment suitable for academic and organizational elections.

Abstract

Traditional voting systems often face issues such as impersonation, vote manipulation, and lack of transparency due to centralized control. To address these problems, this project proposes a secure blockchain-based voting system integrated with face recognition and one-time password (OTP) authentication. The system verifies voter identity using face recognition technology to ensure that only registered users can access the platform. To enhance security, OTP verification is added as a second authentication layer, providing multi-factor security and preventing unauthorized access. After successful authentication, voters can cast their votes through a secure interface. The votes are encrypted and stored on the blockchain as immutable transactions, ensuring transparency, integrity, and protection against tampering. Smart contracts enforce voting rules such as one-person-one-vote and automatically handle vote validation and counting. By combining biometric authentication with blockchain technology, the system provides a secure and reliable voting solution suitable for academic and organizational-level elections.

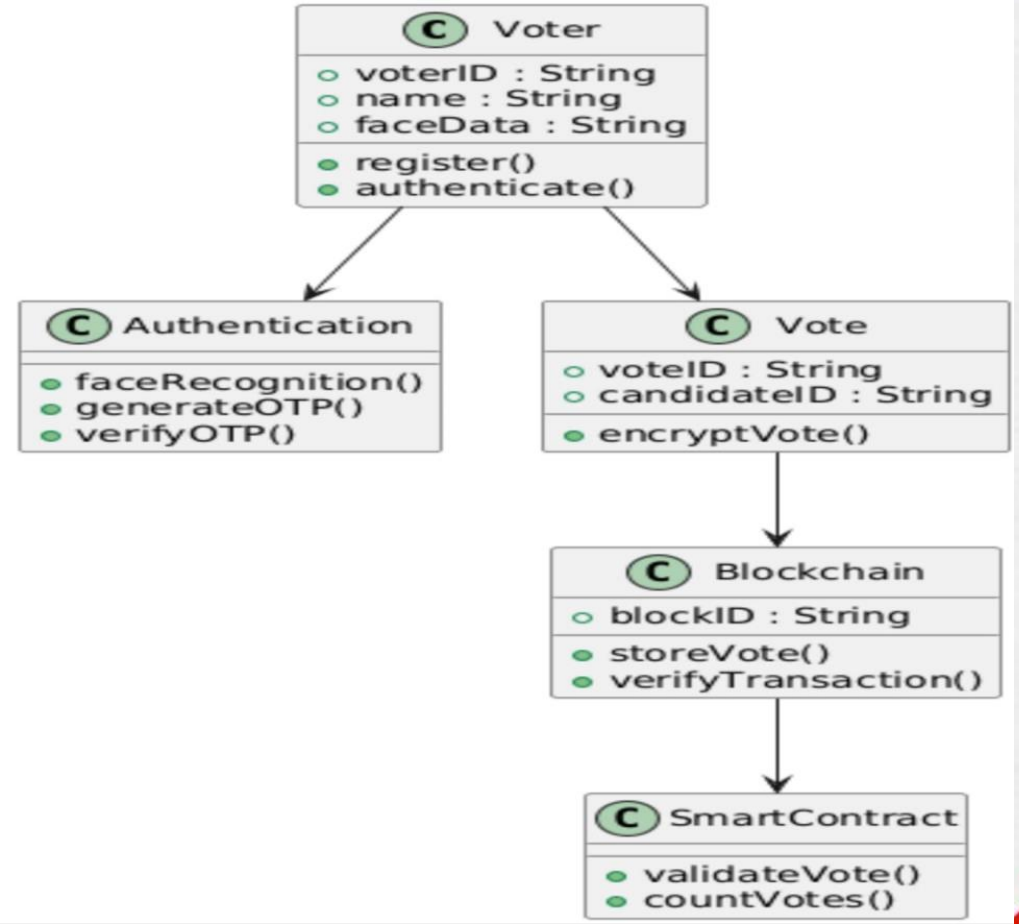
Existing System

The existing voting system typically relies on traditional paper-based methods or centralized electronic voting systems. In paper-based voting, the process requires physical presence, manual verification of voter identity, and manual counting of votes, which can be time-consuming and prone to human errors. Centralized electronic voting systems improve speed but still depend on a single authority to manage voter data and store votes. This centralized control creates risks such as impersonation, data manipulation, vote tampering, and lack of transparency. Additionally, weak authentication mechanisms may allow unauthorized individuals to participate in the voting process. These limitations highlight the need for a more secure, transparent, and tamper-resistant voting system.

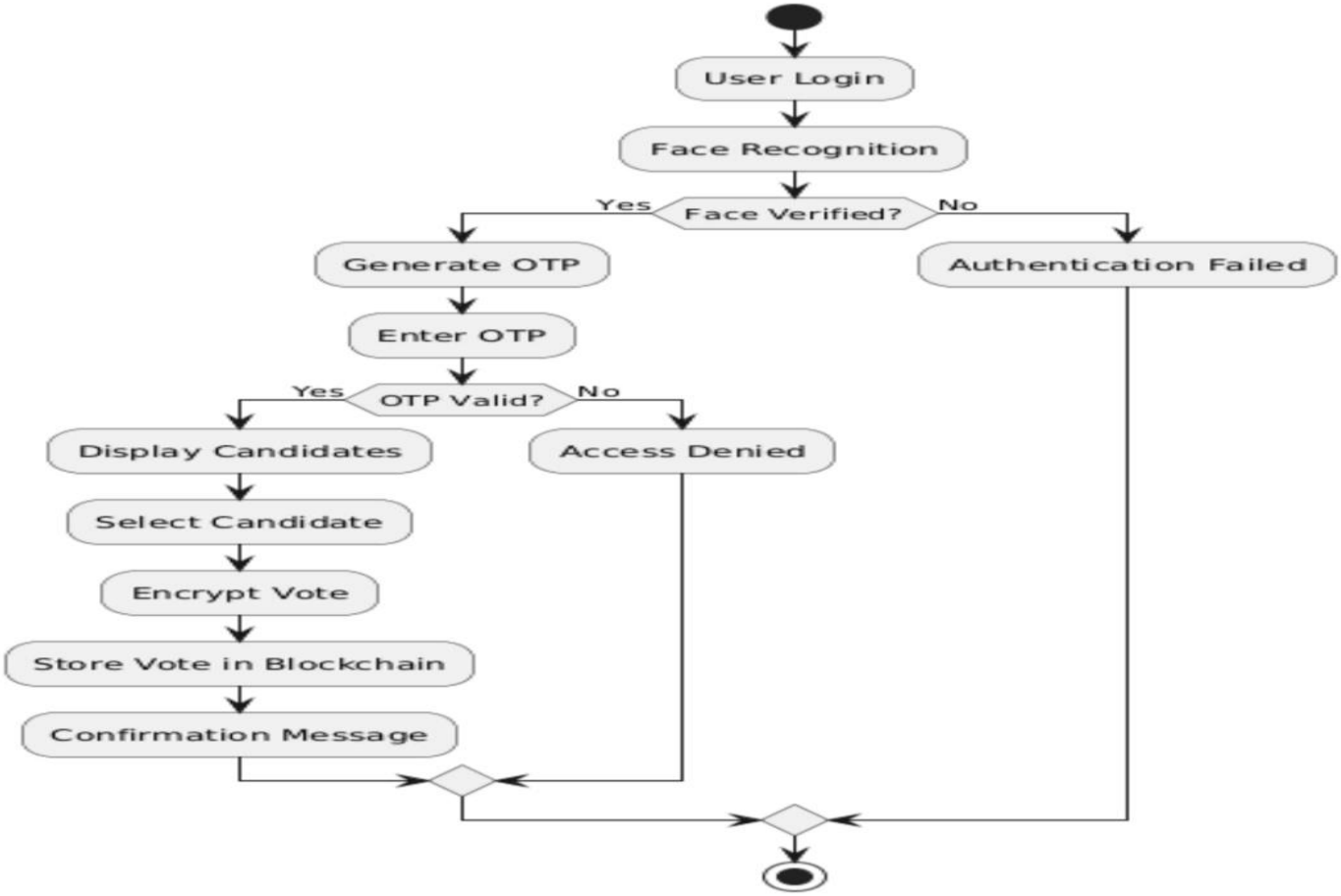
Proposed System

The proposed system introduces a secure and decentralized blockchain-based voting framework designed to enhance transparency and prevent fraud. In this system, voter authentication is performed using face recognition technology to verify the identity of registered users. To strengthen security further, one-time password (OTP) verification is implemented as an additional authentication layer, ensuring multi-factor verification before allowing access to the voting platform. Once the voter is successfully authenticated, they can cast their vote through a secure interface. The selected vote is encrypted and stored on the blockchain as an immutable transaction, preventing any unauthorized modification or tampering. Smart contracts are used to enforce voting rules such as one-person-one-vote and to automate vote validation and counting without manual intervention. By combining biometric authentication with blockchain technology, the proposed system ensures a secure, transparent, and reliable voting process suitable for academic and organizational-level elections.

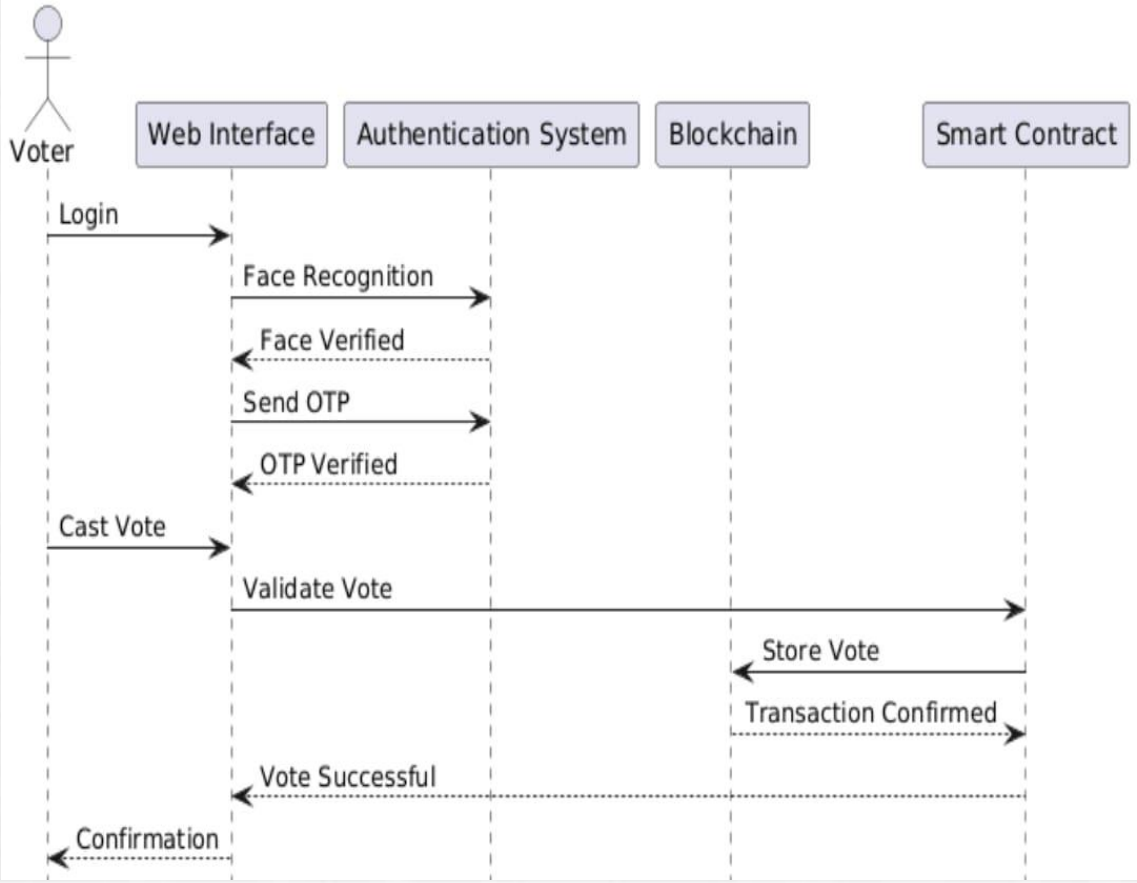
Class Diagram



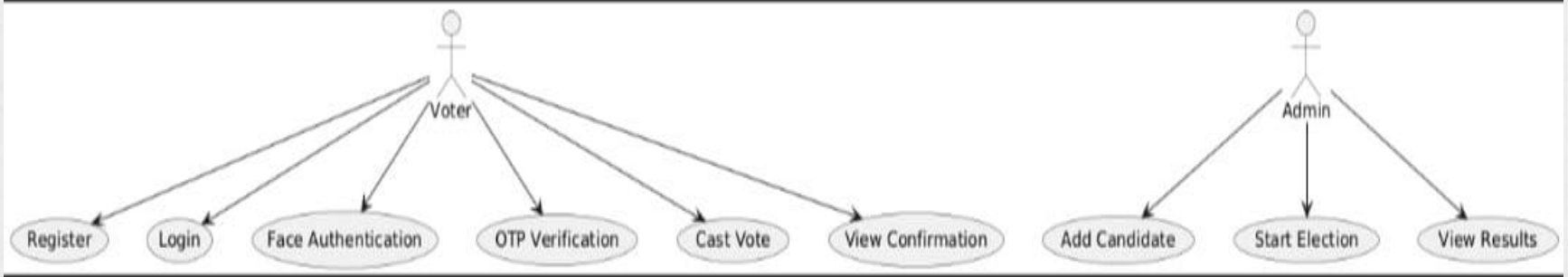
Activity Diagram



Sequence diagram



Use Case Diagram



Hardware and Software Requirements

Hardware:

- Processor: Intel i3 or above
- RAM: 4 GB minimum
- Storage: 256 GBWeb Camera (for face recognition)
- Internet Connection

Software:

- OS: Windows / Linux / macOS
- Programming Language: Python
- Frontend: HTML, CSS, JavaScript
- Blockchain: Ethereum (Ganache)
- Smart Contract: Solidity
- Tools: MetaMask, VS Code
- Libraries: OpenCV, Face Recognition

Conclusion

The proposed blockchain-based voting system provides a secure and transparent solution to the limitations of traditional voting methods. By integrating face recognition and OTP authentication, the system ensures that only authorized and verified users can participate in the voting process. The use of blockchain technology guarantees immutability and protection against vote tampering, while smart contracts enforce rules such as one-person-one-vote and enable automatic vote counting. This combination of biometric authentication and decentralized technology enhances security, trust, and reliability in the voting process. The system is suitable for academic and organizational-level elections and can be further expanded for larger-scale implementations in the future.

THANK YOU